

Azure Migration Planning Guide

Key considerations for migrating applications to Azure

From on-premises datacenters or other cloud platforms

Scope	Audience	Outcome
Application migration from on-premises or another cloud to Azure	Architecture, DevOps, platform, security, and application teams	Clear decisions, migration waves, cutover readiness, and post-migration optimization

Prepared as a practical checklist with flows, decision tables, and migration readiness controls.

1. Executive summary

Migration to Azure should not be treated as a simple infrastructure copy activity. A successful migration defines the target operating model first, then moves workloads in controlled waves with validated security, networking, data, observability, cost, and cutover plans.

- Start with discovery and dependency mapping before selecting the Azure service target.
- Prepare an Azure landing zone for identity, network, governance, monitoring, and security before production migration.
- Classify each workload by migration strategy: rehost, replatform, refactor, rearchitect, replace, or retire.
- Validate database and storage movement early because these usually drive downtime, risk, and rollback complexity.
- Treat cutover and rollback as formal runbooks, not as informal weekend activities.
- Optimize after migration by right-sizing, using managed services, improving security posture, and reducing operational burden.

Migration lifecycle flow

1. Discover	2. Assess	3. Design	4. Pilot	5. Migrate	6. Optimize
Inventory apps, servers, databases, integrations, and dependencies	Check Azure readiness, sizing, compatibility, risk, and cost	Create landing zone, security, network, identity, and target architecture	Move a low-risk workload and validate migration runbooks	Execute migration waves with testing, cutover, and rollback plans	Right-size, modernize, govern, secure, and improve operations

Note: Use migration waves. Avoid moving all applications at once unless the estate is very small and low risk.

2. Migration strategy selection

Strategy	Meaning	Typical Azure target	When to use
Rehost	Lift and shift with minimal change.	Azure VM, Azure VMware Solution	Fast migration, legacy OS/app constraints, short datacenter exit timeline.
Replatform	Small changes to use managed services.	App Service, Azure SQL Managed Instance, Azure Files	Reduce VM operations without major code change.
Refactor	Change application internals or packaging.	Containers, App Service, Azure Container Apps	Improve scalability, deployment speed, and maintainability.
Rearchitect	Redesign application architecture.	AKS, event-driven services, managed databases	Modernization, high availability, cloud-native patterns.
Replace	Move capability to SaaS.	Microsoft 365, Dynamics, third-party SaaS	Commodity business function where custom hosting is not needed.
Retire	Decommission unused workload.	N/A	No business owner, obsolete functionality, or replaced capability.

3. Workload assessment checklist

Assessment area	Key questions	Expected output
Application inventory	What apps exist? Who owns them? Which are business critical?	Application catalog with owners and criticality.
Runtime and OS	Which runtimes, frameworks, OS versions, agents, and libraries are required?	Compatibility and remediation list.
Dependencies	What databases, APIs, file shares, queues, DNS names, certificates, and third-party services are used?	Dependency map and migration sequence.
Performance baseline	What are current CPU, memory, disk IOPS, throughput, latency, and peak patterns?	Right-sizing and performance baseline.
Data profile	How much data exists? What changes frequently? What downtime is acceptable?	Data migration method and cutover window.
Security and compliance	What encryption, audit, regulatory, access, and residency controls apply?	Security control matrix.
SLA, RTO, RPO	What uptime and recovery requirements must Azure meet?	Reliability and DR design.
Cost	What is current cost and expected Azure run cost?	Cost model, budgets, and optimization options.

4. Azure landing zone readiness

Prepare the Azure foundation before migrating production workloads. The landing zone should establish governance, security, connectivity, identity, and operational consistency.

Landing zone area	Considerations	Examples
Subscription structure	Separate environments, business units, or workload types as needed.	Prod, non-prod, shared services, sandbox.
Management groups	Apply policy and access boundaries at scale.	Platform, landing zones, decommissioned.
Resource organization	Standardize resource groups, naming, tags, and ownership.	app-env-region-rg; costCenter, owner, environment.
Policy and governance	Restrict regions, SKUs, public IPs, diagnostic settings, and encryption.	Azure Policy initiatives and exemptions.
Shared services	Centralize logging, security, DNS, firewall, and private connectivity.	Log Analytics, Azure Firewall, Private DNS, Key Vault.
Operations	Define backup, monitoring, alerts, patching, and incident response.	Azure Monitor, Defender for Cloud, Azure Backup.

5. Target architecture decision flow

Hosting model decision flow

Need OS control?	Standard web/API?	Container app?	Kubernetes required?	Event-driven code?	Managed data possible?
Use Azure VM	Use App Service	Use App Service for Containers or	Use AKS	Use Azure	Use Azure managed

		Container Apps		Functions	databases
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Note: Select the simplest managed service that satisfies the application requirements. Use VMs only when OS-level control or legacy constraints require them.

Existing workload	Recommended Azure target	Why	Watch-outs
VM-based legacy app	Azure VM	Minimal change and maximum OS control.	Patch, backup, monitoring, and licensing remain your responsibility.
Web/API application	Azure App Service	Managed runtime, deployment slots, autoscale, easy CI/CD.	Check runtime support, VNet integration, and file system assumptions.
Simple containerized app	Azure Container Apps or App Service for Containers	Container deployment without managing Kubernetes.	Validate networking, scaling model, secrets, and ingress.
Existing Kubernetes app	AKS	Kubernetes compatibility and advanced orchestration.	Requires cluster operations, node pools, ingress, policies, and upgrades.
SQL Server with instance features	Azure SQL Managed Instance	Near SQL Server compatibility with managed operations.	Check unsupported features, sizing, networking, and migration downtime.
Simple relational database	Azure SQL Database, PostgreSQL, or MySQL	Managed database operations and scaling.	Schema, extensions, collation, and performance need validation.
File share dependency	Azure Files	SMB/NFS options and managed file shares.	Latency, permissions, locking behavior, and throughput limits.
Object storage	Azure Blob Storage	Durable, scalable object storage.	Access model, lifecycle rules, naming, and application SDK changes.

6. Identity and access management

- Define whether applications will use Microsoft Entra ID, hybrid identity, or an existing external identity provider.
- Use Azure RBAC through groups rather than assigning permissions directly to individuals.
- Use managed identities for Azure service-to-service authentication wherever possible.
- Store secrets, keys, and certificates in Azure Key Vault, not in code repositories or VM files.
- Plan privileged access using least privilege, just-in-time access, and approval workflows for production.

7. Network and connectivity design

Network topic	Questions to answer	Design decision
Topology	Hub-spoke, Virtual WAN, or isolated VNets?	Define standard network pattern before migration.

Network topic	Questions to answer	Design decision
Hybrid connectivity	VPN or ExpressRoute? Bandwidth and latency needs?	Choose connectivity and routing model.
IP addressing	Any overlapping CIDR ranges across on-prem, Azure, or other clouds?	Avoid overlaps; redesign before migration if needed.
DNS	How will private DNS, conditional forwarding, and service discovery work?	Define DNS resolution path.
Ingress	Public endpoint, Application Gateway, Front Door, or private access?	Choose secure entry pattern.
Egress	Direct internet, firewall inspection, NAT Gateway, or forced tunneling?	Control outbound traffic and audit paths.
Private access	Which PaaS services require private endpoints?	Reduce public exposure for data and critical services.

8. Data, database, and storage migration

Data migration often determines the actual complexity, downtime, and rollback feasibility. Validate data paths early, not at the end of the project.

Area	Considerations	Migration options	Validation
Databases	Schema compatibility, stored procedures, extensions, collation, size, replication, downtime.	Backup/restore, online migration, replication, Database Migration Service.	Schema diff, row counts, performance tests, application smoke tests.
File shares	SMB/NFS, permissions, locking, latency, throughput, user access.	AzCopy, Azure File Sync, robocopy, Data Box.	Access tests, checksums, permission validation.
Object storage	Buckets/containers, metadata, lifecycle, access policies, SDK differences.	AzCopy, storage migration tools, custom sync.	Object counts, size totals, metadata and application reads.
Backups	Retention, immutability, restore testing, compliance.	Azure Backup, Recovery Services Vault, native database backups.	Restore drills and RPO verification.

9. Security, compliance, and risk controls

Control area	Key controls	Common Azure services
Secrets and keys	Centralized secret storage, certificate lifecycle, key rotation.	Azure Key Vault, managed identities.
Network security	Segmentation, private endpoints, firewall rules, WAF, DDoS protection.	NSG, Azure Firewall, Application Gateway WAF, Front Door.
Threat protection	Vulnerability management, workload protection, posture management.	Microsoft Defender for Cloud.
Logging and audit	Central logs, retention, immutable evidence, access reviews.	Azure Monitor, Log Analytics, Microsoft Sentinel.
Data protection	Encryption at rest and in transit, backup, retention, data residency.	Storage encryption, database TDE, Azure Backup.

Control area	Key controls	Common Azure services
Governance	Policy enforcement, allowed locations, allowed SKUs, tagging, public exposure control.	Azure Policy, management groups, blueprints/initiatives.

10. Reliability, availability, and disaster recovery

- Define service-level targets before selecting Azure SKUs or regions.
- Capture RTO and RPO for each application and database.
- Use Availability Zones where the service and region support them.
- Design DR separately from backup. Backup protects data; DR restores service capability.
- Test restores and failover procedures before declaring the migration complete.

Requirement	Azure design option	Best for	Caution
Single-region high availability	Availability Zones, zone-redundant PaaS, load balancers.	Most production workloads.	Check zone support by service and region.
Multi-region resilience	Front Door, geo-replication, paired regions, active/passive or active/active.	Critical services with regional outage tolerance.	Operational complexity and cost increase.
VM disaster recovery	Azure Site Recovery.	Lift-and-shift VM workloads.	Failover testing and app consistency are required.
Data recovery	Automated backups, geo-redundant backups, point-in-time restore.	Databases and storage.	Verify RPO and restore time through testing.

11. DevOps, deployment, and release management

Target	CI/CD approach	Artifact	Typical tools
Azure VM	Build, copy package or files, run deployment script.	Build output or package.	GitHub Actions, SSH, Azure Run Command, Ansible.
Container on VM	Build image, push to ACR, pull and restart container on VM.	Docker image.	GitHub Actions, ACR, Docker.
App Service	Build package and deploy to web app or deployment slot.	ZIP/JAR/WAR/folder.	GitHub Actions, azure/webapps-deploy.
AKS	Build image, push to ACR, deploy manifest or Helm chart.	Docker image and Kubernetes manifests.	GitHub Actions, ACR, kubectl, Helm.
Infrastructure	Provision and update consistently through code.	IaC modules/templates.	Terraform, Bicep, ARM templates.

12. Observability and operations

- Enable application logs, platform logs, metrics, traces, and diagnostics before cutover.
- Create dashboards for business-critical metrics and technical health indicators.

- Configure alerts for availability, latency, error rate, CPU, memory, storage, certificate expiry, and database health.
- Prepare operational runbooks for restart, rollback, scale-out, restore, and incident response.
- Define ownership for after-hours support, escalation, and post-cutover hypercare.

13. Cost and licensing

Cost area	What to consider	Optimization actions
Compute	VM size, App Service plan, AKS node pools, autoscale.	Right-size, schedule non-prod shutdown, use reserved capacity or savings plans.
Storage	Disk type, snapshots, backup retention, archive tiers.	Use lifecycle policies and avoid overprovisioned premium storage.
Network	Egress, NAT Gateway, firewall, cross-region transfer.	Reduce unnecessary cross-zone/region traffic and monitor egress.
Monitoring	Log ingestion and retention can become expensive.	Filter noisy logs, set retention by compliance need.
Licensing	Windows Server, SQL Server, third-party software.	Use Azure Hybrid Benefit where eligible and review vendor licensing.

14. Cloud-to-cloud migration considerations

Source cloud concept	Azure equivalent	Migration note
AWS EC2 / GCP Compute Engine	Azure VM	Validate OS images, disks, networking, and licensing.
AWS EKS / GKE	AKS	Review ingress, storage classes, node pools, IAM/RBAC, and monitoring.
AWS ECS/Fargate	Azure Container Apps or AKS	Map task definitions, environment variables, secrets, scaling, and ingress.
AWS S3 / GCS	Azure Blob Storage	Plan object migration, metadata, lifecycle policies, and SDK changes.
AWS RDS / Cloud SQL	Azure SQL, PostgreSQL, MySQL, or SQL Managed Instance	Validate engine compatibility and downtime strategy.
CloudWatch / Cloud Logging	Azure Monitor and Log Analytics	Translate alerts, dashboards, and retention policies.
Secrets Manager / Secret Manager	Azure Key Vault	Update application secret retrieval and access policies.

15. On-premises migration considerations

- Inventory VMware, Hyper-V, physical servers, databases, file shares, and network appliances.
- Plan Active Directory, DNS, domain join, group policies, and hybrid identity carefully.
- Validate VPN or ExpressRoute connectivity before testing application migration.
- Check legacy OS versions, unsupported runtimes, hardware-bound licensing, and static IP assumptions.

- Replace datacenter backup, monitoring, patching, firewall, and DR processes with Azure-native or approved enterprise equivalents.
- Use a staged approach: rehost where necessary, stabilize, then modernize to PaaS or containers.

16. Cutover and rollback plan

Production cutover flow

Prepare	Pre-sync	Freeze	Final sync	Switch	Validate	Hypercare
Freeze window, approvals, runbooks, backups	Seed data and reduce DNS TTL	Stop writes or enter maintenance mode	Copy deltas and verify consistency	Update DNS, routing, or traffic manager	Smoke tests, logs, metrics, business signoff	Monitor closely and keep rollback window open

Note: Rollback must include data handling. After users write data to Azure, rollback to the old platform may require reverse synchronization.

Cutover readiness checklist	Status
Application owner and business validator identified	Not started / In progress / Done
Final migration runbook approved	Not started / In progress / Done
Rollback trigger and steps documented	Not started / In progress / Done
Backups completed and restore tested	Not started / In progress / Done
Data sync method tested	Not started / In progress / Done
DNS TTL reduced in advance	Not started / In progress / Done
Monitoring dashboards and alerts enabled	Not started / In progress / Done
Security controls validated	Not started / In progress / Done
Performance smoke tests prepared	Not started / In progress / Done
Communication plan ready	Not started / In progress / Done
Post-cutover support window confirmed	Not started / In progress / Done

17. Final migration approval checklist

Category	Approval question	Owner
Business	Has the business owner approved downtime and validation criteria?	Application owner
Architecture	Is the Azure target architecture approved and documented?	Solution architect
Security	Are identity, secrets, network, encryption, and audit controls approved?	Security team
Platform	Is landing zone, network, monitoring, backup, and policy configuration complete?	Platform team
Data	Has database/storage migration been tested and reconciled?	Database/data owner
Operations	Are runbooks, alerts, dashboards, and escalation paths ready?	Operations team
Cost	Is expected Azure cost reviewed and budgeted?	FinOps / owner
Cutover	Are cutover and rollback plans tested and approved?	Migration lead

18. References

- [Microsoft Cloud Adoption Framework for Azure - Migration](#)
- [Azure Migrate documentation](#)
- [Azure landing zones documentation](#)
- [Azure Well-Architected Framework](#)
- [Azure Architecture Center](#)